

Circles, Arcs and Sectors

A sector is a fraction of a circle: $\frac{\theta}{360}$ of the area and of the circumference. Perimeter = arc + two radii.



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THE TOPIC ON ONE MAP

Circles, Arcs and Sectors

01 Circumference and area

$\pi d, \pi r^2$ circles in a square

02 Arc length and sector area

fraction $\frac{\theta}{360}$ perimeter of a sector

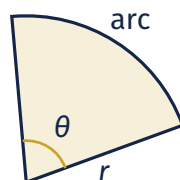
03 Segments and shaded regions

sector - triangle region common to two circles

04 Working backwards

find r from the area find x

THE SECTOR One picture, three formulas:



$$\text{arc length} = \frac{\theta}{360} \times 2\pi r$$

$$\text{sector area} = \frac{\theta}{360} \times \pi r^2$$

$$\text{perimeter} = \text{arc} + 2r$$

01 Circumference and Area of Circles

Circumference = $\pi d = 2\pi r$; area = πr^2 . Read carefully: diameter or radius? Semicircles and quarter circles are fractions of these.

WHEN YOU SEE → YOU DO

“diameter 140 cm”

→ $C = \pi \times 140$; for area halve it first

“in terms of π ”

→ leave π in the answer

circles inside a square

→ the square side = diameters

“perimeter” of a shape with curves

→ curved parts + every straight edge

WORKED MODEL · JUNE 2024 P2HR

A circle inside a square of area 81 cm^2 . Total wire for the square and the circle?

1 side = 9, square perimeter = 36

square

2 circle diameter = 9: 9π

circle

ANSWER $36 + 9\pi = 64.3 \text{ cm}$

IDEA 01

Practise: Circumference and Area of Circles

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Both formulas

CH 4 · TOPIC 6 2 marks

A circle has radius 5 cm. Find its circumference and area in terms of π .

SELECTED PRACTICE

Q2 PRACTICE Practice · Easy

CH 4 · TOPIC 6 2 marks

1 A circle has a diameter of 140 cm.

Work out the circumference of the circle. Give your answer correct to 3 significant figures.

(2 marks)

Q3**PRACTICE**

Practice · Medium

CH 4 · TOPIC 6 3 marks

- 2 The diagram shows a circle drawn inside a square.

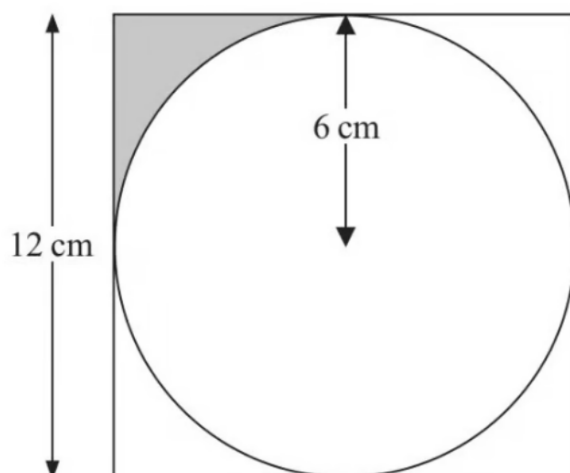


Diagram **NOT**
accurately drawn

The circle has a radius of 6 cm.

The square has a side of length 12 cm.

Work out the shaded area.

Give your answer in terms of π .

(3 marks)

Q4**PRACTICE**

Practice · Hard

CH 4 · TOPIC 6

5 marks

- 1 The diagram shows a plan of Brian's lawn.

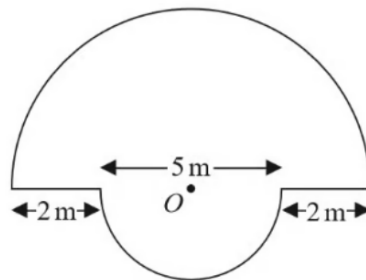


Diagram **NOT**
accurately drawn

The edge of the lawn consists of two semicircles and two straight lines.

Each semicircle has centre O .

The diameters of the semicircles are 9 m and 5 m.

Brian is going to put lawn edging around the edge of the lawn. Lawn edging is sold in 2.4 metre rolls.

Brian has £35

Lawn edging

£3.99 per roll or 3 rolls for £10

Has Brian got enough money to buy all the rolls of lawn edging he needs?

You must show all your working.

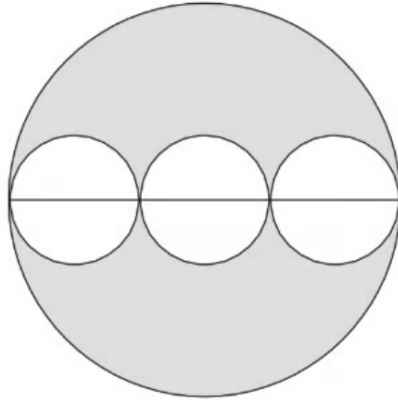
(5 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 6 **3 marks**

- 18** Three identical small circles are drawn inside one large circle, as shown in the diagram. The centres of the small circles lie on the diameter of the large circle.



Find the fraction of the large circle that is shaded.

(3 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 1H · Q8

CH 4 · TOPIC 6 3 marks

- 8 The region, shown shaded in the diagram, is a path.

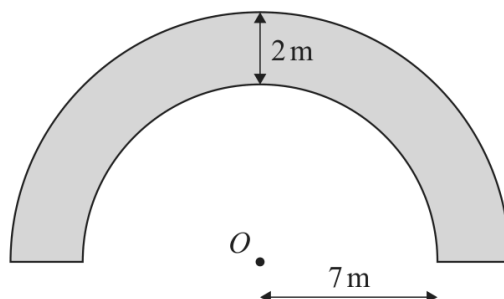


Diagram **NOT**
accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O , and two straight lines.

The inner semicircle has a radius of 7 metres.

The path has a width of 2 metres.

Work out the perimeter of the path.

Give your answer correct to one decimal place.

..... m

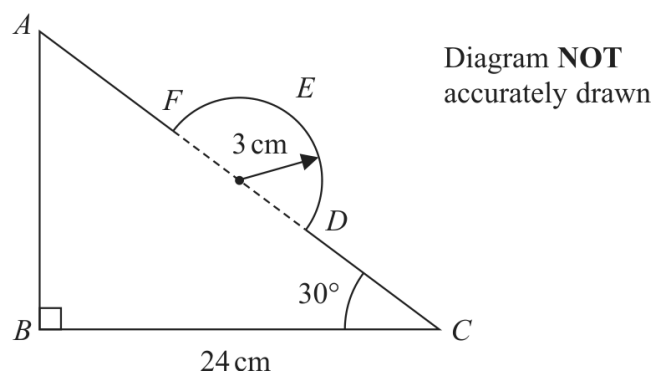
(Total for Question 8 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q9

CH 4 · TOPIC 6 5 marks

- 9 In the diagram, ABC is a right-angled triangle and DEF is a semicircular arc.



In triangle ABC

$$BC = 24 \text{ cm}$$

$$\text{angle } ABC = 90^\circ$$

$$\text{angle } BCA = 30^\circ$$

The points D and F lie on AC so that DF is the diameter of the semicircular arc DEF
 The radius of the semicircular arc is 3 cm.

Work out the length of $AFEDC$

Give your answer correct to 2 significant figures.

..... cm

(Total for Question 9 is 5 marks)

Q8 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 2H · Q20**CH 4 · TOPIC 6** **4 marks**

20 The diagram shows four identical circles drawn inside a square.

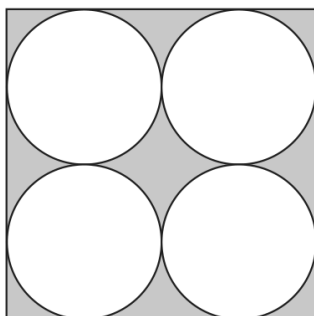


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1HR · Q4

CH 4 · TOPIC 6 5 marks

- 4 The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

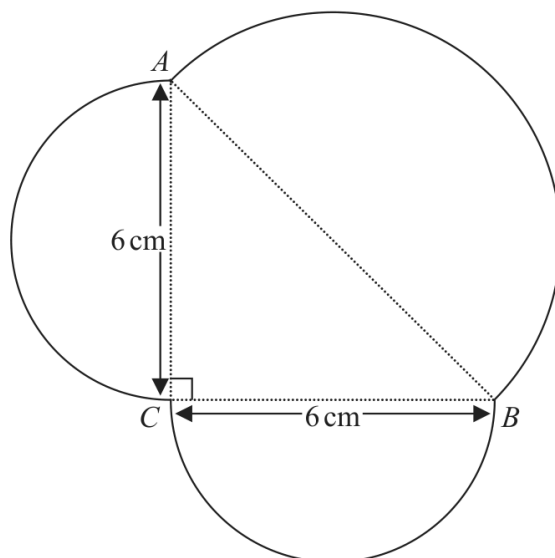


Diagram **NOT**
accurately drawn

AB , BC and CA are each the diameter of a semicircle.

$BC = CA = 6$ cm.

Work out the perimeter of the shape.

Give your answer correct to one decimal place.

..... cm

(Total for Question 4 is 5 marks)

Q10**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q5

CH 4 · TOPIC 6 3 marks

- 5 The diagram shows a shaded shape made from three identical semicircles, *X*, *Y* and *Z*

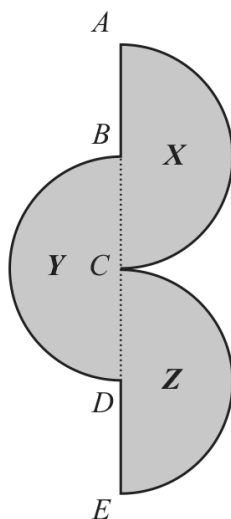


Diagram **NOT**
accurately drawn

ABCDE is a straight line.

AC is the diameter of semicircle *X* and *B* is the centre of semicircle *X*

BD is the diameter of semicircle *Y* and *C* is the centre of semicircle *Y*

CE is the diameter of semicircle *Z* and *D* is the centre of semicircle *Z*

$$AC = BD = CE = 20 \text{ cm}$$

Work out the perimeter of the shaded shape.

Give your answer correct to the nearest whole number.

..... cm

(Total for Question 5 is 3 marks)

Q11**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q5

CH 4 · TOPIC 6 4 marks

- 5 The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square $ABCD$

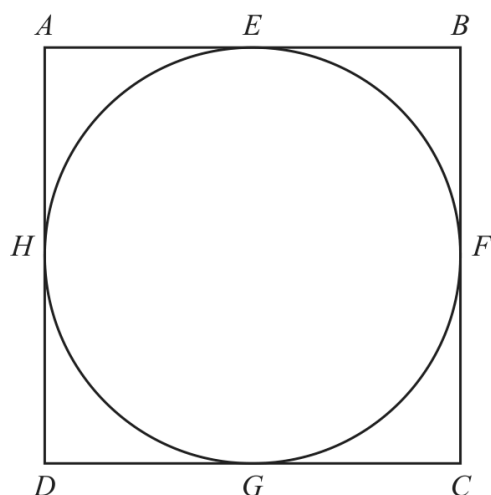


Diagram **NOT**
accurately drawn

The circle touches the square at the points E, F, G and H

The area of the square is 81 cm^2

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 5 is 4 marks)

Q12**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2H · Q21

CH 4 · TOPIC 6 5 marks

21 The diagram shows a square $ABCD$ and a circle.

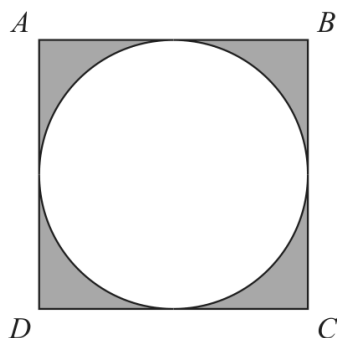


Diagram **NOT**
accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of AC

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 21 is 5 marks)

Q13**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q6

CH 4 · TOPIC 6 5 marks

- 6 A large circle with centre O contains 3 identical small circles.
Each circle touches two other circles.

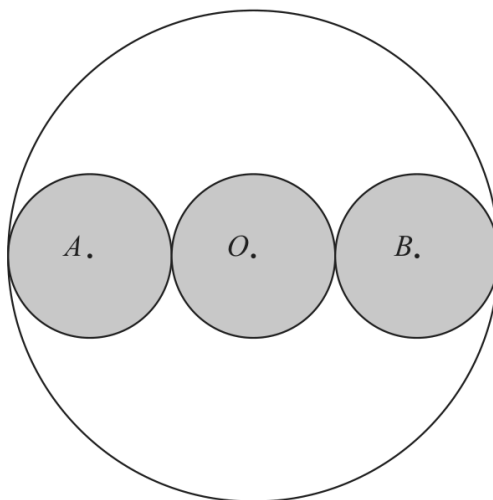


Diagram **NOT**
accurately drawn

A , O and B are the centres of the small circles.
 AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 6 is 5 marks)

02 Arc Length and Sector Area

Find the fraction $\frac{\theta}{360}$ first. Multiply it by the full circumference for the arc, by the full area for the sector.

WHEN YOU SEE → YOU DO

“length of the arc”

$$\rightarrow \frac{\theta}{360} \times 2\pi r$$

“area of the sector”

$$\rightarrow \frac{\theta}{360} \times \pi r^2$$

“perimeter of the sector”

$$\rightarrow \text{arc} + 2r \text{ (do not forget the radii)}$$

a reflex sector (e.g. $360 - 40$)

$$\rightarrow \text{use the bigger angle}$$

Fraction $\frac{\theta}{360}$

$\times 2\pi r$ or $\times \pi r^2$

Add radii for perimeter

WORKED MODEL • JANUARY 2021 P2HR

Radius 9 cm; the shaded sector is the circle minus 40° . Perimeter?

$$1 \quad \frac{320}{360} \times 2\pi(9) = 16\pi$$

arc

ANSWER $16\pi + 18 = 68.3 \text{ cm}$

× **TRAP** Giving only the arc when asked for the perimeter. Add the two radii.

IDEA 02

Practise: Arc Length and Sector Area

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Fraction of the circle

CH 4 · TOPIC 6 3 marks

A sector has radius 9 cm and angle 80° . Find its arc length and area to 3 s.f.

SELECTED PRACTICE**Q15** PRACTICE Practice · Easy

CH 4 · TOPIC 6 2 marks

3

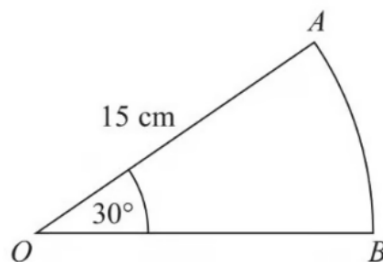


Diagram **NOT**
accurately drawn

OAB is a sector of a circle, centre O . The radius of the circle is 15 cm. The angle of the sector is 30° .

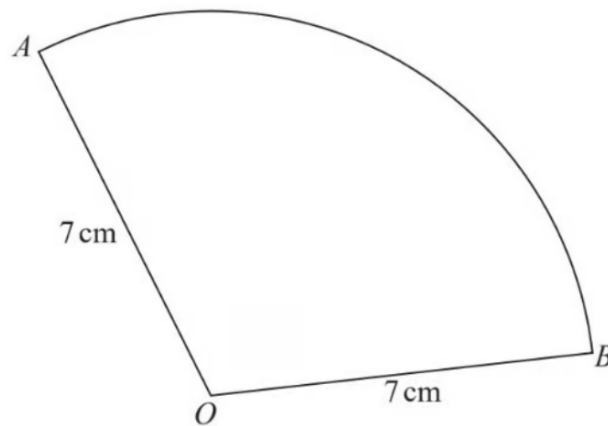
Calculate the area of sector OAB . Give your answer correct to 3 significant figures.

(2 marks)

Q16 PRACTICE Practice · Medium

CH 4 · TOPIC 6 4 marks

- 6 OAB is a sector of a circle with centre O and radius 7 cm.



The area of the sector is 40 cm^2 .

Calculate the perimeter of the sector.

Give your answer correct to 3 significant figures.

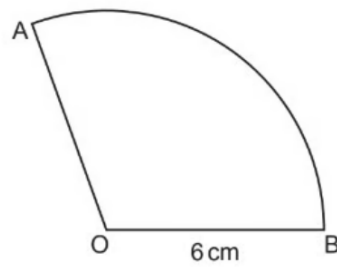
(4 marks)

Q17**PRACTICE**

Practice · Hard

CH 4 · TOPIC 6 5 marks

- 11 AOB is a sector of a circle, centre O and radius 6cm. The length of arc AB is 5π cm.



Not to scale

Find the area of the sector.

Give your answer in terms of π .

..... cm^2
(5 marks)

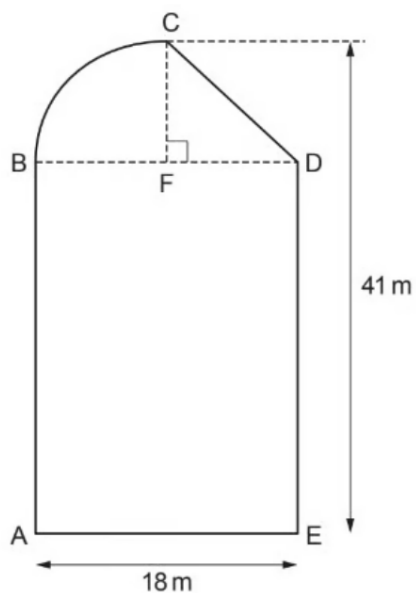
Q18**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 6 6 marks

15 The diagram shows a shape ABCDE.

The shape is made from a rectangle, a right-angled triangle and a quarter of a circle.



F is the mid-point of BD.

AE = 18 m and the perpendicular distance from C to AE is 41 m.

Work out the **perimeter** of the shape ABCDE.

..... m

(6 marks)

PAST PAPERS

Q19 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 2HR · Q11

CH 4 · TOPIC 6 4 marks

11 A circle centre O has radius 9 cm.

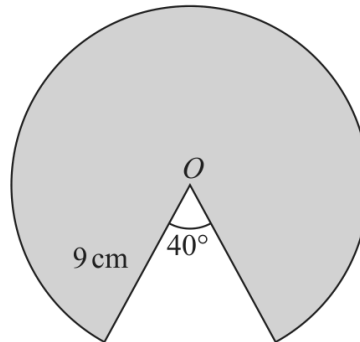


Diagram **NOT**
accurately drawn

Calculate the perimeter of the shaded sector of the circle.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 11 is 4 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q12

CH 4 · TOPIC 6 2 marks

12 The diagram shows a sector of a circle with radius 7 cm.

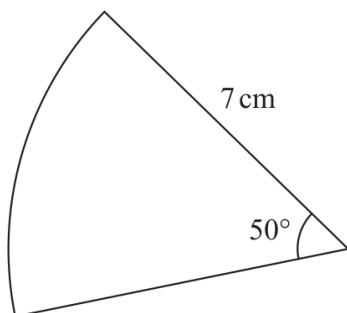


Diagram **NOT**
accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

(Total for Question 12 is 2 marks)

Q21**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q11

CH 4 · TOPIC 6 2 marks

- 11 The diagram shows sector OPQ of a circle, centre O

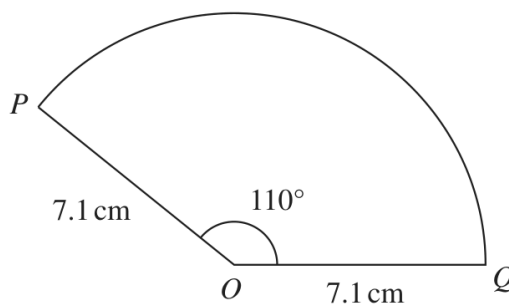


Diagram **NOT**
accurately drawn

$$OP = OQ = 7.1\text{ cm}$$

$$\text{Angle } POQ = 110^\circ$$

Calculate the area of sector OPQ

Give your answer correct to one decimal place.

..... cm^2

(Total for Question 11 is 2 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q13

CH 4 · TOPIC 6 3 marks

13 The diagram shows a sector AOB of a circle with centre O

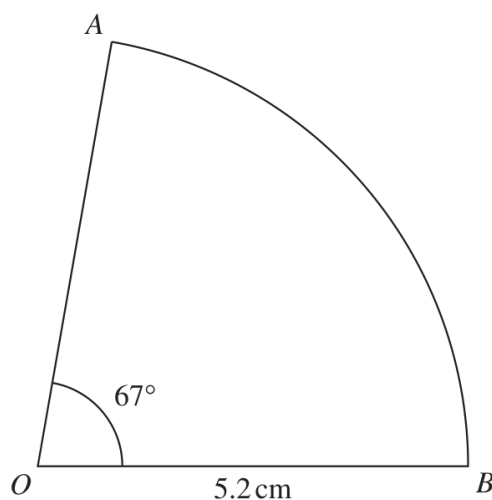


Diagram **NOT**
accurately drawn

Angle $AOB = 67^\circ$
 $OA = OB = 5.2 \text{ cm}$

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 13 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q11

CH 4 · TOPIC 6 2 marks

11 $OABC$ is a sector of a circle, centre O

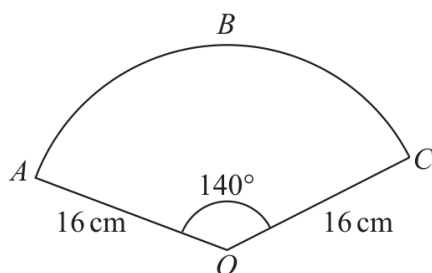


Diagram **NOT**
accurately drawn

$$\text{angle } AOC = 140^\circ$$

$$OA = OC = 16 \text{ cm}$$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 2 marks)

Q24**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q15

CH 4 · TOPIC 6 2 marks

15 AOB is a sector of a circle with centre O and radius 12 cm

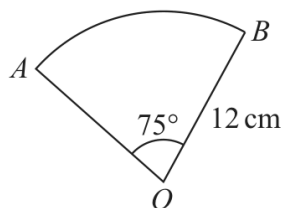


Diagram **NOT**
accurately drawn

Work out the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 15 is 2 marks)

03 Segments and Shaded Regions

Segment = sector – triangle ($\frac{1}{2}r^2 \sin \theta$). Any shaded region: whole shape minus the unshaded parts.

WHEN YOU SEE → YOU DO

“segment”

→ sector – $\frac{1}{2}r^2 \sin \theta$

“region common to both circles”

→ two segments added

circle inside a triangle / square

→ shape area – circle area

“shaded = unshaded”

→ each is half the total

ZONE SECTOR

$$\frac{\theta}{360} \pi r^2$$

ZONE TRIANGLE

$$\frac{1}{2} r^2 \sin \theta$$

ZONE SEGMENT

sector – triangle

WORKED MODEL • NOVEMBER 2023 P2H

A circle inside an equilateral triangle of side 10 cm. Shaded area outside the circle?

1 triangle = $\frac{1}{2}(10)^2 \sin 60^\circ = 25\sqrt{3}$

triangle

2 radius = $\frac{5}{\sqrt{3}}$: circle = $\frac{25\pi}{3}$

inscribed circle

ANSWER $25\sqrt{3} - \frac{25\pi}{3} = 17.1 \text{ cm}^2$

IDEA 03

Practise: Segments and Shaded Regions

My questions first, then selected practice, then real past papers.

Q25 DR ESLAM Segment

CH 4 · TOPIC 6 3 marks

A sector has radius 6 cm and angle 90° . Find the area of the segment cut off by the chord.

SELECTED PRACTICE

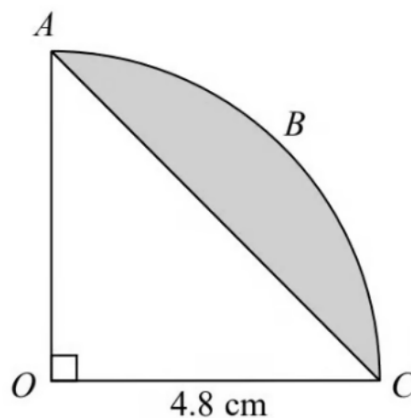
Q26

PRACTICE

Practice · Hard

CH 4 · TOPIC 6 3 marks

6



The arc ABC is a quarter of a circle with centre O and radius 4.8 cm . AC is a chord of the circle.

Work out the area of the shaded segment. Give your answer correct to 3 significant figures.

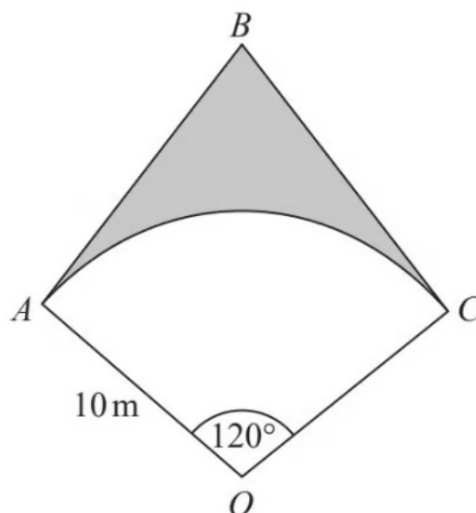
(3 marks)

Q27**PRACTICE**

Practice · Hard

CH 4 · TOPIC 6 5 marks

9



OAC is a sector of a circle, centre O , radius 10 m .

BA is the tangent to the circle at point A .

BC is the tangent to the circle at point C .

Angle $AOC = 120^\circ$

Calculate the area of the shaded region.

Give your answer correct to 3 significant figures.

(5 marks)

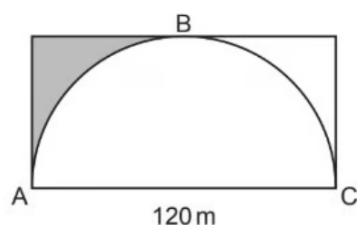
Q28**PRACTICE**

Practice · Hard

CH 4 · TOPIC 6 5 marks

- 12 The diagram shows a semi-circle inside a rectangle of length 120 m. The semi-circle touches the rectangle at A, B and C.

Not to scale



Calculate the **perimeter** of the shaded region.
Give your answer correct to 3 significant figures.

..... m

(5 marks)

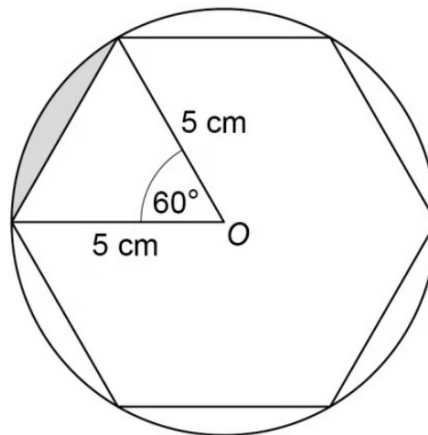
Q29**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 6 4 marks

- 9 The vertices of a regular hexagon lie on a circle with centre O and radius 5 cm

Not drawn accurately



Work out the shaded area.

Give your answer in the form $\frac{a\pi - b\sqrt{c}}{12}$ where a , b and c are integers.

(4 marks)

PAST PAPERS

Q30 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 2HR · Q20

CH 4 · TOPIC 6 5 marks

20 A , B and C are points on a circle with centre O .

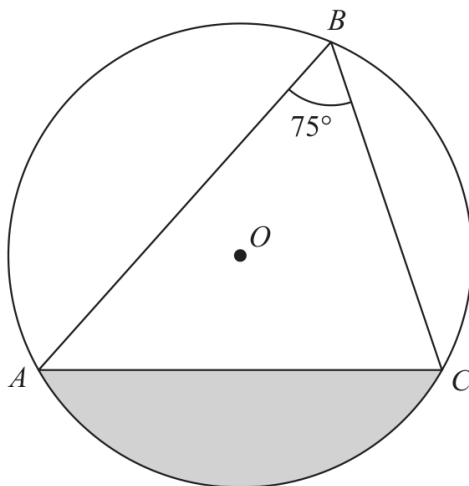


Diagram **NOT**
accurately drawn

Angle $ABC = 75^\circ$

The area of the shaded segment is 200 cm^2

Calculate the radius of the circle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Q31 PAST PAPER Edexcel 4MA1 · May 2021 · Paper 1H · Q25

CH 4 · TOPIC 6 6 marks

- 25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

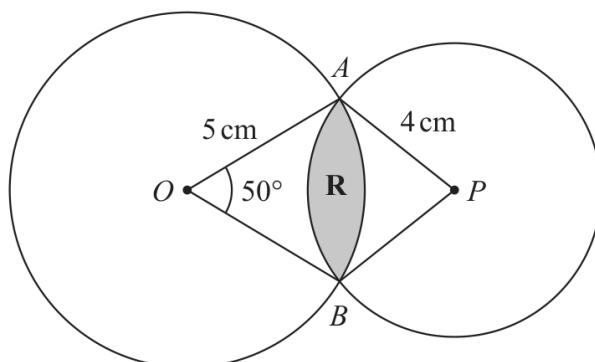


Diagram **NOT**
accurately drawn

One of the circles has centre O and radius 5 cm .

The other circle has centre P and radius 4 cm .

Angle $AOB = 50^\circ$

Calculate the area of region **R**.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 25 is 6 marks)

Q32

PAST PAPER

Edexcel 4MA1 · May 2023 · Paper 1H · Q22

CH 4 · TOPIC 6 6 marks

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

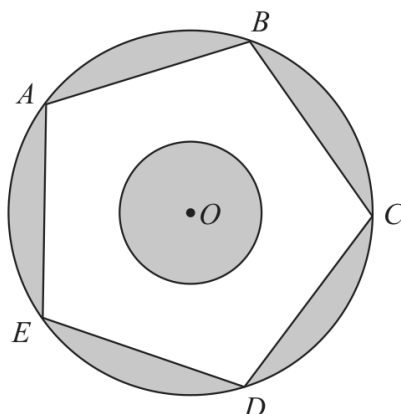


Diagram **NOT**
accurately drawn

A, B, C, D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

shaded area = unshaded area

Work out the radius of the smaller circle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 2H · Q20

CH 4 · TOPIC 6 4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.

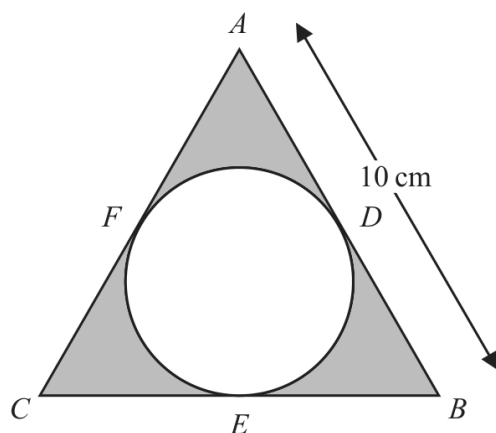


Diagram **NOT**
accurately drawn

D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 20 is 4 marks)

Q34 PAST PAPER Edexcel 4MA1 · November 2024 · Paper 2H · Q25

CH 4 · TOPIC 6 5 marks

25 The diagram shows an equilateral triangle ABC and a circle with centre O

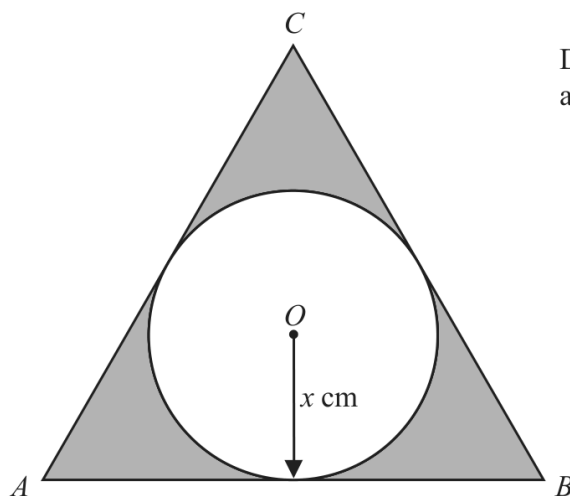


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

$n = \dots\dots\dots$

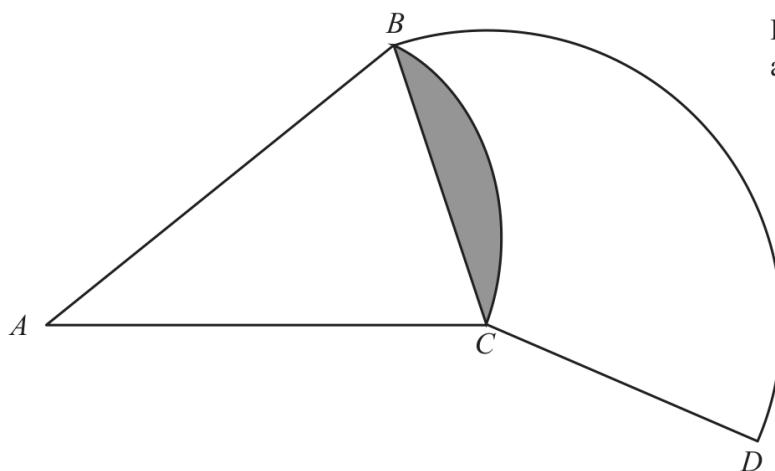
(Total for Question 25 is 5 marks)

Q35**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q25

CH 4 · TOPIC 6 6 marks

25

Diagram **NOT**
accurately drawn BAC is a sector of a circle, centre A BCD is a sector of a circle, centre C Angle $BAC = 40^\circ$ Angle $BCD = 130^\circ$ Area of shaded segment = 28 cm^2 Find the length of the arc BD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 25 is 6 marks)

04 Working Backwards

The length or area is given: write the formula with the unknown, set it equal, solve. For a segment you may need to try values or use the calculator's solve.

WHEN YOU SEE → YOU DO

“arc length is 5 cm, angle 55°”

→ $\frac{55}{360}2\pi r = 5$, solve for r

“area of sector S is twice circle C ”

→ write both, one equation in r

“find P in the form $r(c\pi + k)$ ”

→ arc $+2r$, factor out r

“area of triangle OAB is 120”

→ $\frac{1}{2}r^2 \sin \theta = 120$ gives r

WORKED MODEL · JUNE 2022 P1H

Sector S : radius $(r + 7)$, 45° . Circle C : radius $(r - 2)$. $S = 2C$. Find r .

1 $\frac{1}{8}\pi(r + 7)^2 = 2\pi(r - 2)^2$

set up

2 $(r + 7)^2 = 16(r - 2)^2 \Rightarrow r + 7 = 4(r - 2)$

square root

ANSWER $r = 5$

IDEA 04

Practise: Working Backwards

My questions first, then selected practice, then real past papers.

Q36 DR ESLAM Backwards

CH 4 · TOPIC 6 3 marks

A sector of angle 60° has area $24\pi \text{ cm}^2$. Find the radius.

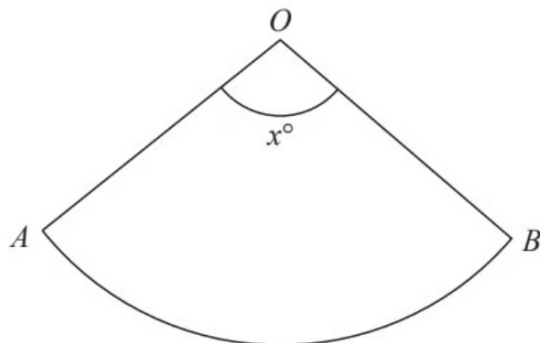
SELECTED PRACTICE

Q37**PRACTICE**

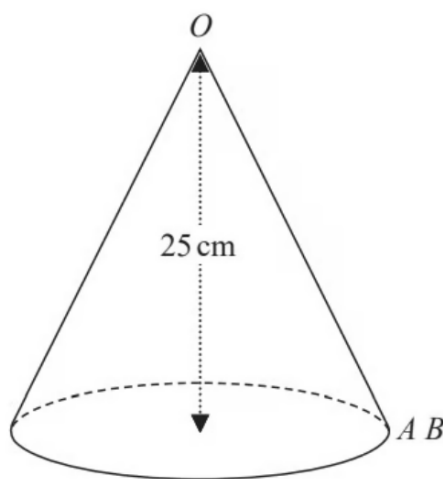
Practice · Very Hard

CH 4 · TOPIC 6 6 marks

- 6 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

Diagram NOT
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

Diagram NOT
accurately drawn

The height of the cone is 25 cm.

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

$x = \dots\dots\dots$

(6 marks)

Q38

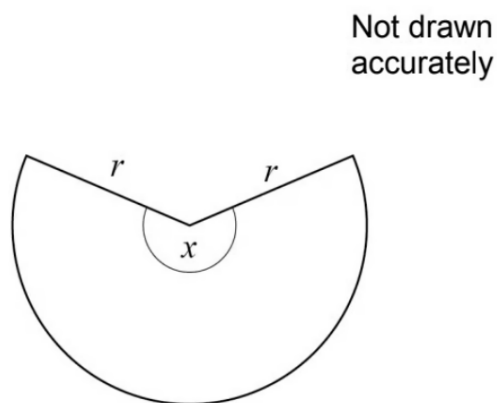
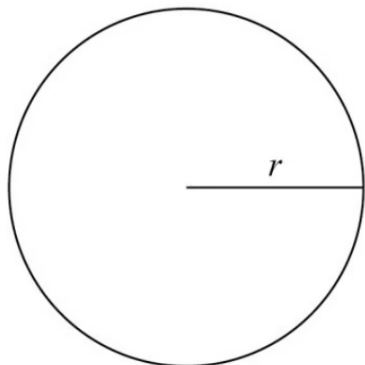
PRACTICE

Practice · Very Hard

CH 4 · TOPIC 6 4 marks

10 Here are a circle and a sector of the circle.

They each have radius r .



circumference of circle = perimeter of sector

Work out the size of angle x .

Give your answer in terms of π .

.....degrees

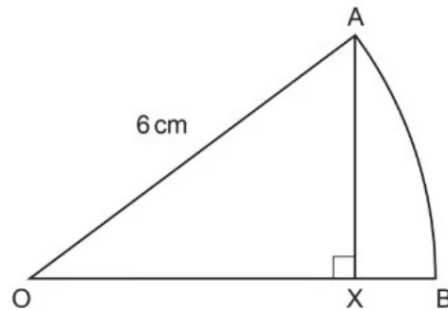
(4 marks)

Q39**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 6 6 marks

- 16** OAB is a sector of a circle, centre O.
OA = 6 cm and AX is perpendicular to OB.



The area of sector OAB is $6\pi\text{cm}^2$.

Show that $AX = 3\sqrt{3}$ cm.

(6 marks)

PAST PAPERS

Q40**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2H · Q22

CH 4 · TOPIC 6 6 marks

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

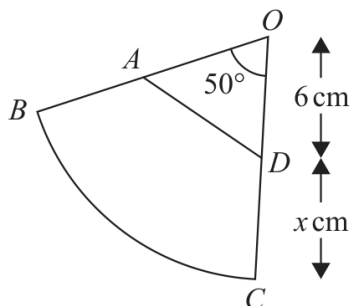


Diagram **NOT**
accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$

(Total for Question 22 is 6 marks)

Q41**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q16

CH 4 · TOPIC 6 4 marks

16 The diagram shows a circle with centre O

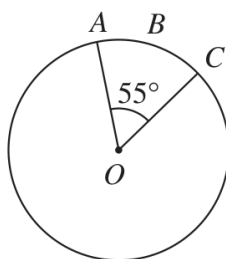


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

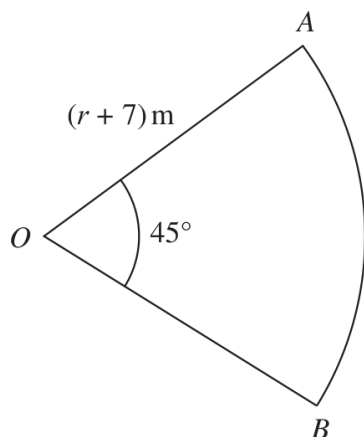
..... cm^2

(Total for Question 16 is 4 marks)

Q42 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1H · Q19

CH 4 · TOPIC 6 5 marks

19

Diagram **NOT**
accurately drawn

OAB is a sector **S** of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45^\circ$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r

Show your working clearly.

Question 19 continued.

 $r = \dots\dots\dots$

(Total for Question 19 is 5 marks)

Q43**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q13

CH 4 · TOPIC 6 3 marks

13 OAB is a sector of a circle with centre O and radius r cm.

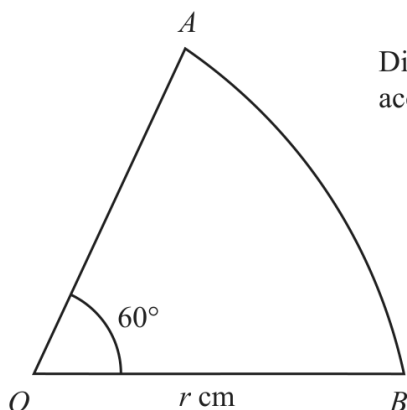


Diagram **NOT**
accurately drawn

Angle $AOB = 60^\circ$

The perimeter of the sector is P cm.

Find a formula for P in terms of r

Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

(Total for Question 13 is 3 marks)

Q44 **PAST PAPER** Edexcel 4MA1 · May 2024 · Paper 1H · Q20**CH 4 · TOPIC 6** **4 marks**

20 The diagram shows a sector OAC of a circle centre O

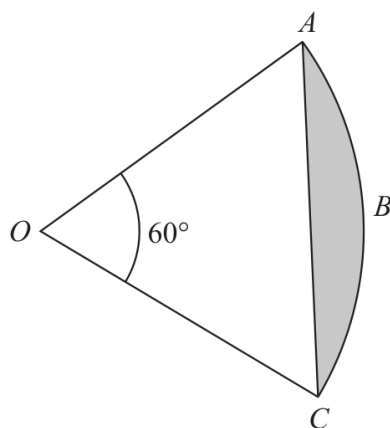


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

..... cm

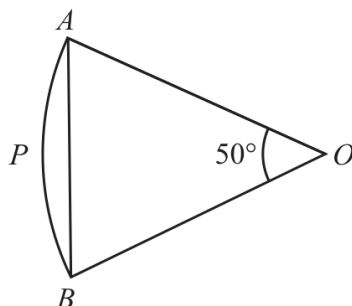
(Total for Question 20 is 4 marks)

Q45**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q16

CH 4 · TOPIC 6 4 marks

16 $OAPB$ is a sector of a circle, centre O



Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 16 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------|---|
| 01 | Circles | $\pi d, \pi r^2$; radius or diameter? |
| 02 | Sectors | $\frac{\theta}{360}$ of $2\pi r$ or πr^2 ; perimeter adds $2r$. |
| 03 | Segments | Sector $-\frac{1}{2}r^2 \sin \theta$; shaded = whole - unshaded. |
| 04 | Backwards | Formula = value, solve for r or θ . |

BOTTOM LINE Fraction of the circle → arc or area → perimeter: add the radii.

Answers

Circles, Arcs and Sectors • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CIRCUMFERENCE AND AREA OF CIRCLES

Q1	Dr Eslam • Both formulas	10π cm; 25π cm ²
Q2	Practice • Easy	440 cm
Q3	Practice • Medium	$144 - 36\pi$ cm ²
Q4	Practice • Hard	edging 26.0 m: 11 rolls, cheapest £37.98 > £35: No
Q5	Practice • Very Hard	$1 - 3 \times \frac{1}{9} = \frac{2}{3}$
Q6	Edexcel 4MA1 • January 2021 • Paper 1H • Q8	$7\pi + 9\pi + 4 = 54.3$ m
Q7	Edexcel 4MA1 • January 2022 • Paper 2HR • Q9	$AC - 6 + 3\pi = 31$ cm
Q8	Edexcel 4MA1 • June 2022 • Paper 2H • Q20	side 13.65: 54.6 cm
Q9	Edexcel 4MA1 • May 2023 • Paper 1HR • Q4	$6\pi + 3\sqrt{2}\pi = 32.2$ cm
Q10	Edexcel 4MA1 • November 2023 • Paper 1H • Q5	$30\pi + 40 = 134$ cm
Q11	Edexcel 4MA1 • June 2024 • Paper 2HR • Q5	$36 + 9\pi = 64.3$ cm
Q12	Edexcel 4MA1 • June 2024 • Paper 2H • Q21	side 19.3: $AC = 27.3$ cm
Q13	Edexcel 4MA1 • November 2025 • Paper 2H • Q6	large radius 25.46, small radius 8.49: 226 cm ²

IDEA 02 · ARC LENGTH AND SECTOR AREA

- Q14** Dr Eslam · Fraction of the circle
arc 12.6 cm; area 56.5 cm^2
- Q15** Practice · Easy
 58.9 cm^2
- Q16** Practice · Medium
25.4 cm
- Q17** Practice · Hard
angle 150° : $15\pi \text{ cm}^2$
- Q18** Practice · Very Hard
 $32 + 14.1 + 12.7 + 32 + 18 = 108.9 \text{ m}$
- Q19** Edexcel 4MA1 · January 2021 · Paper 2HR · Q11
 $16\pi + 18 = 68.3 \text{ cm}$
- Q20** Edexcel 4MA1 · May 2021 · Paper 1H · Q12
6.1 cm
- Q21** Edexcel 4MA1 · January 2022 · Paper 1HR · Q11
 48.4 cm^2
- Q22** Edexcel 4MA1 · June 2022 · Paper 2HR · Q13
16.5 cm
- Q23** Edexcel 4MA1 · June 2025 · Paper 1H · Q11
 313 cm^2
- Q24** Edexcel 4MA1 · November 2025 · Paper 1H · Q15
 94.2 cm^2

IDEA 03 · SEGMENTS AND SHADED REGIONS

Q25	Dr Eslam • Segment	$9\pi - 18 = 10.3 \text{ cm}^2$
Q26	Practice • Hard	$5.76\pi - 11.52 = 6.58 \text{ cm}^2$
Q27	Practice • Hard	$100\sqrt{3} - \frac{100\pi}{3} = 68.5 \text{ m}^2$
Q28	Practice • Hard	$60 + 60 + 30\pi = 214 \text{ m}$
Q29	Practice • Very Hard	$\frac{50\pi - 75\sqrt{3}}{12}$
Q30	Edexcel 4MA1 • January 2021 • Paper 2HR • Q20	$r = 13.7 \text{ cm}$
Q31	Edexcel 4MA1 • May 2021 • Paper 1H • Q25	3.06 cm^2
Q32	Edexcel 4MA1 • May 2023 • Paper 1H • Q22	$r = 3.45 \text{ cm}$
Q33	Edexcel 4MA1 • November 2023 • Paper 2H • Q20	17.1 cm^2
Q34	Edexcel 4MA1 • November 2024 • Paper 2H • Q25	$n = 3\sqrt{3} - \pi = 2.05$
Q35	Edexcel 4MA1 • June 2025 • Paper 1HR • Q25	$AC = 31.8: \text{ arc } BD = 49.4 \text{ cm}$

IDEA 04 • WORKING BACKWARDS

- Q36** Dr Eslam · Backwards $\frac{1}{6}\pi r^2 = 24\pi: r = 12$
- Q37** Practice · Very Hard cone $r = 7.82$, slant $26.2: x = 107$
- Q38** Practice · Very Hard $2\pi r = 2r + \frac{x}{360}2\pi r: x = 360 - \frac{360}{\pi} = 245.4^\circ$
- Q39** Practice · Very Hard $\frac{\theta}{360}36\pi = 6\pi: \theta = 60^\circ, AX = 6 \sin 60^\circ = 3\sqrt{3}$
- Q40** Edexcel 4MA1 · January 2021 · Paper 2H · Q22 $x = 5.89$
- Q41** Edexcel 4MA1 · January 2022 · Paper 1H · Q16 $r = 5.21: 85.2 \text{ cm}^2$
- Q42** Edexcel 4MA1 · June 2022 · Paper 1H · Q19 $r = 5$
- Q43** Edexcel 4MA1 · June 2024 · Paper 2HR · Q13 $P = r\left(\frac{1}{3}\pi + 2\right)$
- Q44** Edexcel 4MA1 · May 2024 · Paper 1H · Q20 $r = 19.1: 41.9 \text{ cm}$
- Q45** Edexcel 4MA1 · November 2024 · Paper 1H · Q16 $r = 17.7: 137 \text{ cm}^2$